

Myhill-Nerode Theorem

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August 28, 2024

1 Recap: Equivalence Relations

Recall the definition of an equivalence relation.

Definition 1 (Equivalence Relations). Suppose S is a set. A (binary) relation R on S is said to be an *equivalence relation* if and only if

- R is reflexive: xRx , for all $x \in S$;
- R is symmetric: xRy implies yRx , for all $x, y \in S$;
- R is transitive: xRy and yRz implies xRz , for all $x, y, z \in S$.

Note that the set S is called the *domain* of the relation R .

Example 1. Suppose $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$ and $\sim$ is defined by

$$x \sim y \text{ if and only if } x \bmod 3 = y \bmod 3,$$

for all $x, y \in S$. Thus, for example, $1 \sim 4$ and $1 \sim 7$ but $1 \not\sim 2$ and $1 \not\sim 0$.

It is easy to check that $\sim$ is an equivalence relation.

Based on a given equivalence relation R , if we group elements in the domain that are related in R together, we can see that the relation R partitions the domain into a number of subsets of elements, called *equivalence classes*.

Definition 2 (Equivalence Classes). Given an equivalence relation R on S , for each element $x \in S$, the equivalence class in R containing x , denoted by $[x]^R$, is defined by

$$[x]^R = \{y \in S \mid xRy\}.$$

When it is clear which relation we are referring to, we shall omit the superscript R from $[x]^R$ and simply write the equivalence class containing x by $[x]$.

Proposition 1. Suppose R is an equivalence relation on S . For all $x, y \in S$,

- $x \in [x]$,
- $x \in [y]$ if and only if $y \in [x]$,

- xRy if and only if $[x] = [y]$.

Proof. This can shown easily from the properties of equivalence relations. $\square$

Example 2. Recall the equivalence relation $\sim$ from the previous example. It is easy to see that there are 3 distinct equivalence classes, namely

- $[0] = [3] = [6] = \{0, 3, 6\};$
- $[1] = [4] = [7] = \{1, 4, 7\};$
- $[2] = [5] = [8] = \{2, 5, 8\}.$

Definition 3 (Finite Index). An equivalence relation R is said to be *of finite index* if and only if there are finitely many equivalence classes in R .

Given an equivalence relation R of finite index, the index of R , sometimes denoted by $\#R$, is the number of distinct equivalence classes of R .

Thus the relation $\sim$ defined in the previous example is of finite index, as it has 3 distinct equivalence classes.

2 Myhill-Nerode Theorem

For the rest of this document, let Σ denote a non-empty alphabet and Σ^* the set of all (finite) strings over Σ .

Definition 4 (Myhill-Nerode Relations). Given a language L , a *Myhill-Nerode relation* for L is any (binary) relation $\equiv$ on Σ^* such that all of the following are true:

- $\equiv$ is an equivalence relation.
- $\equiv$ is a *right congruence* with respect to concatenation, i.e. for any strings $x, y \in \Sigma^*$ and character $a \in \Sigma$,

$$\text{if } x \equiv y, \text{ then } (x \circ a) \equiv (y \circ a).$$

- $\equiv$ refines L , i.e. for any string $x, y \in \Sigma^*$,

$$\text{if } x \equiv y, \text{ then } x \in L \iff y \in L.$$

- $\equiv$ is of finite index.

Example 3. Suppose $\Sigma = \{0, 1\}$ and $L = \{x \in \Sigma^* \mid x \text{ contains an odd number of } 1\}$. We can show that there is a Myhill-Nerode relation for L .

First, let say that two integers have the *same parity* if and only if they are either both odd or both even.

Define the relation $\equiv$ on Σ^* as follows: for any strings $x, y \in \Sigma^*$,

$$\begin{aligned} & x \equiv y \\ & \text{if and only if} \\ & \text{the number of } 1 \text{ in } x \text{ and the number of } 1 \text{ in } y \text{ have the same parity.} \end{aligned}$$

Thus, for instance, $110 \equiv 010111$ because the numbers of 1 in these two strings are both even, but $110 \not\equiv 01011$ because the number of 1 in the first string is even but the number of 1 in the latter string is odd.

It is straightforward to check that $\equiv$ is an equivalence relation with two equivalence classes, one containing all the strings with an even number of 1, and the other containing all the strings with an odd number of 1. The language L contains precisely the strings in the latter equivalence class. This implies that $\equiv$ refines L .

We can also show that $\equiv$ is a right congruence (Can you?). Thus, $\equiv$ is a Myhill-Nerode relation for L .

A given language L might or might not have a Myhill-Nerode relation. One famous result is that a Myhill-Nerode relation for L exists if and only if L is a regular language.

Theorem 1 (Myhill-Nerode Theorem). *Suppose L is a language. L is regular if and only if there exists a Myhill-Nerode relation for L .*

Proof. (Proof idea) If L is regular, there is a DFA M recognizing L . We can define a Myhill-Nerode relation for L from the DFA M .

Conversely, if L has a Myhill-Nerode relation $\equiv$, then we can construct a DFA M recognizing L where each equivalence class of $\equiv$ becomes a state in M . $\square$

Myhill-Nerode Theorem can be used to prove that a language is not regular, as we shall show in the proposition below. First, let us prove a useful property of a right congruence.

Lemma 1. *Suppose $\equiv$ is an equivalence relation on Σ^* which is a right congruence, i.e. for any strings $x, y \in \Sigma^*$ and character $a \in \Sigma$,*

$$x \equiv y \text{ implies } (x \circ a) \equiv (y \circ a).$$

Then, for any strings $x, y, z \in \Sigma^$,*

$$x \equiv y \text{ implies } (x \circ z) \equiv (y \circ z).$$

Proof. We shall prove by induction on the length of string z that, for any strings x and y ,

$$x \equiv y \text{ implies } (x \circ z) \equiv (y \circ z). \quad (1)$$

(Basis step) In case the length of z is 0. Statement (1) is obviously true.

(Inductive step) Suppose the hypothesis is true for any string z of length n , where $n \geq 0$. We shall prove that it also holds for any string of length $n + 1$. Let z' be a string of length $n + 1$. Suppose $z' = z \circ a$, where $z \in \Sigma^*$ and $a \in \Sigma$. Then, for any strings x and y ,

$$\begin{aligned} x \equiv y \text{ implies } (x \circ z) &\equiv (y \circ z) && \text{(induction hypothesis)} \\ \text{implies } (x \circ z) \circ a &\equiv (y \circ z) \circ a && \text{(def. of right congruence)} \\ \text{implies } x \circ (z \circ a) &\equiv y \circ (z \circ a) && \text{(associativity of } \circ) \\ \text{implies } x \circ z' &\equiv y \circ z'. \end{aligned}$$

This implies that the induction hypothesis also holds for any string of length $n + 1$.

Thus follows the Lemma. $\square$

To apply Myhill-Nerode Theorem to show that a language L is *not* regular, we can proceed as follows:

1. Assume that L is regular and hence has a Myhill-Nerode relation, say $\equiv$.
2. Find infinitely many (distinct) strings, $x_1, x_2, x_3, \dots$ such that $x_i \not\equiv x_j$ for any $i \neq j$. To show that $x_i \not\equiv x_j$, we only need to find just one string z such that $x_i \circ z \in L$ but $x_j \circ z \notin L$, or vice versa.
3. Conclude from Step 2 that the strings $x_1, x_2, x_3, \dots$ are all in different equivalence classes, which would mean that there are infinitely many equivalence classes. This contradicts the fact that $\equiv$ is a Myhill-Nerode relation, which must be of finite index. Therefore, L is not regular.

We illustrate this in the following examples. Let Σ be $\{0, 1\}$.

Example 4. Show that the language $L = \{0^k 1^k \mid k \geq 0\}$ is *not* regular.

Assume that L is regular. Hence, by Myhill-Nerode Theorem, there must be a Myhill-Nerode relation, say $\equiv$, for L . Consider the strings

$$\epsilon, 0, 00, 000, 0000, 00000, \dots, 0^n, \dots$$

We can show that these strings are all in different equivalence classes. Consider the strings 0^i and 0^j where $i \neq j$. If we append these two strings by 1^i , we have $0^i 1^i \in L$ but $0^j 1^i \notin L$. This implies that $0^i \not\equiv 0^j$. Hence, the strings above are all in different classes, contradicting the fact that $\equiv$ is a Myhill-Nerode relation, which must be of finite index. Therefore, L is *not* regular.

Example 5. Show that the language $L = \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$ is *not* regular.

Assume that L is regular. Hence, by Myhill-Nerode Theorem, there must be a Myhill-Nerode relation, say $\equiv$, for L . Consider the strings

$$1, 01, 001, 0001, 00001, \dots, 0^n 1, \dots$$

We can show that these strings are all in different equivalence classes. Consider the strings $0^i 1$ and $0^j 1$ where $i \neq j$. If we append these two strings by 0^i , we have $0^i 1 0^i$, which is a palindrome, but $0^j 1 0^i$ is *not* a palindrome. This implies that $0^i 1 \not\equiv 0^j 1$. Hence, the strings above are all in different classes, contradicting the fact that $\equiv$ is a Myhill-Nerode relation, which must be of finite index. Therefore, L is *not* regular.

Example 6. Show that the language $L = \{1^{i^2} \mid i \geq 0\}$ is *not* regular.

Assume that L is regular. Hence, by Myhill-Nerode Theorem, there must be a Myhill-Nerode relation, say $\equiv$, for L . Consider the strings

$$1^{0^2}, 1^{1^2}, 1^{2^2}, 1^{3^2}, 1^{4^2}, \dots, 1^{n^2}, \dots$$

We can show that these strings are all in different equivalence classes. Consider the strings 1^{i^2} and 1^{j^2} where $0 \leq i < j$. If we append these two strings by 1^{2i+1} , we have

$$\begin{aligned} 1^{i^2} 1^{2i+1} &= 1^{i^2+2i+1} = 1^{(i+1)^2} \\ 1^{j^2} 1^{2i+1} &= 1^{j^2+2i+1} \end{aligned}$$

The first string $1^{(i+1)^2}$ is in L . We will show that the second string 1^{j^2+2i+1} is *not* in L .

Since $0 < 2i + 1$ and $i < j$, we have

$$\begin{aligned} j^2 &< j^2 + 2i + 1 < j^2 + 2j + 1 \\ j^2 &< j^2 + 2i + 1 < (j + 1)^2. \end{aligned}$$

Since $j^2 + 2i + 1$ is strictly between j^2 and $(j + 1)^2$, it follows that $j^2 + 2i + 1 \neq k^2$ for all integers k . This implies 1^{j^2+2i+1} is *not* in L . Hence, $1^{i^2} \not\equiv 1^{j^2}$. Therefore, the strings in the above list are all in different classes, contradicting the fact that $\equiv$ is a Myhill-Nerode relation, which must be of finite index. Therefore, L is *not* regular.